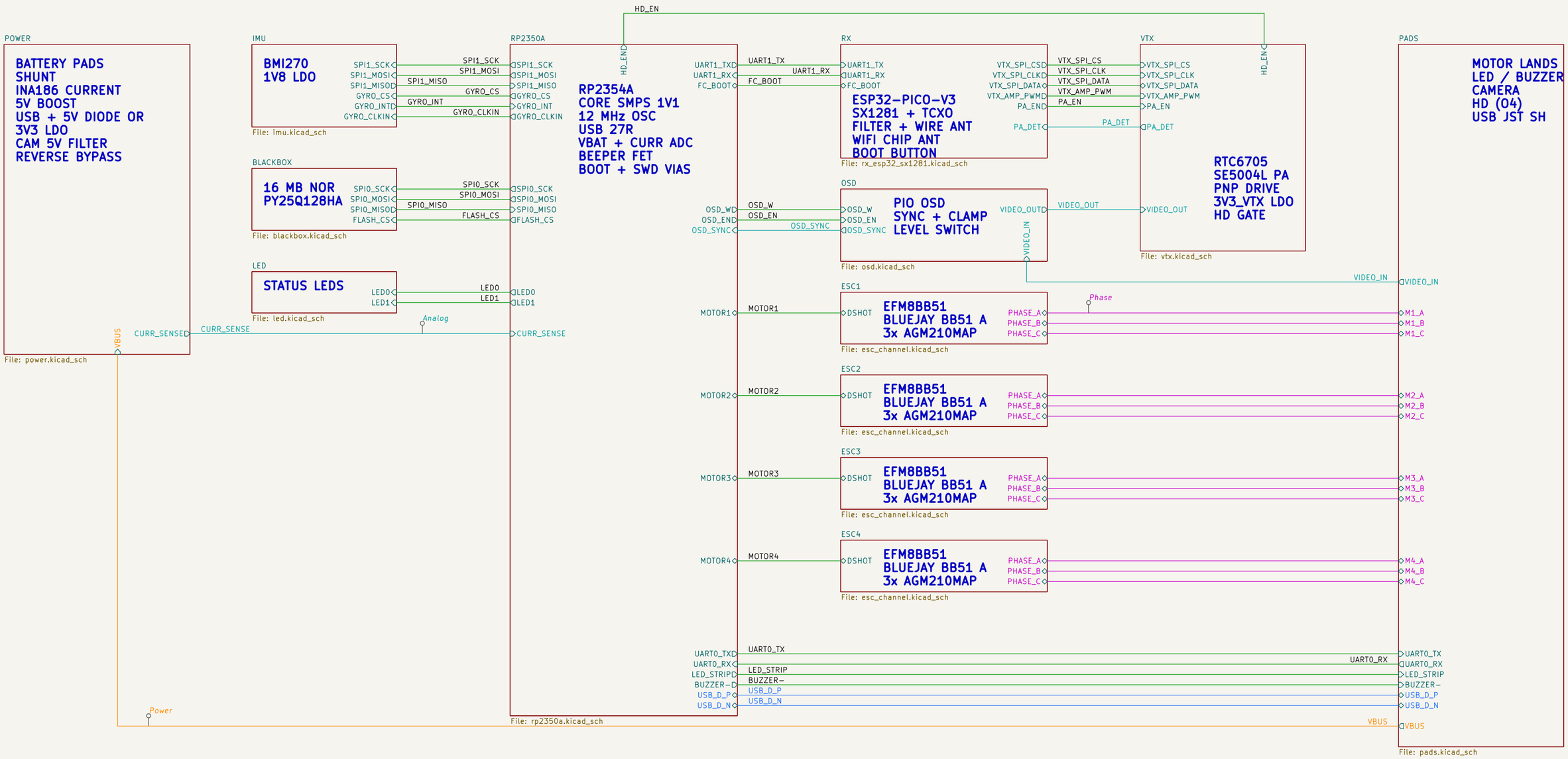


POWER TABLE

Rail	Source	Nominal	Estimated load	Feeds
CSA+	battery pads J2 / J3 (1S)	3.0–4.35 V	4x ESC + 2.0 A board	pad bulk, shunt R2
+BATT	shunt R2 0.5 mOhm	cell	ESC + boost in 1.4–3.0 A	ESC1–4, boost, VBAT ADC, SE5004L VCC
/VBUS	USB J31 pin 4 (hier. net)	5 V	0.42 A, USB only	D9 to +5V
VBUS –> +5V	D9 Schottky	5 V	as VBUS	+5V diode OR
+5V_BST	U2 TPS61022 boost	5.15 V	0.70 A analog / 1.48 A HD (BEC 1.5 A cont.)	D10 to +5V, VHD pad (O4)
+5V	D9 / D10 diode–OR	5 V	0.70 A analog, about 0.25 A HD	3V3 LDOs, CAM filter, 5V pads
+5V_CAM	FB1 ferrite from +5V	5 V	0.12 A	camera pad + plug
(VHD on +5V_BST)	no switch (D114)	5 V	1.20 A O4 Lite	VHD pad
+3.3V	U4 TLV75733	3.3 V	0.20 A (0.48 A Wi–Fi bench)	FC, IMU, NOR, OSD, RX, LEDs, INA186
+3.3V_VTX	U21 LP5912–3.3 (off in HD)	3.3 V	0.10 A typ, about 0.17 A at max PA	RTC6705, PA drive, PA ref
+1.8V_GYRO	U12 TPS7A2018	1.8 V	< 1 mA	BMI270
+1.1V	RP2354A core SMPS	1.1 V	internal	RP2354A DVDD
PA_VCC	+BATT direct (D114)	cell	0.65 A at 27 dBm	SE5004L PA (local, VTX)
PA_VREF	U22 TPS7A20285 (EN = PA_EN, off in HD)	2.85 V	about 10 mA	SE5004L VREF/EN (local, VTX)
GND	battery –	0 V	–	all sheets

Root labels are the project net names. Load estimates require prototype verification.
ESC1–4 = M1 rear right, M2 front right, M3 rear left, M4 front left (Betaflight quad–X).

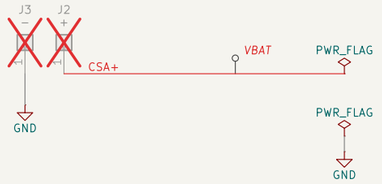
OpenDrone

Sheet: /
File: OpenAIO–Whoop.kicad_sch

Title: OpenAIO–Whoop overview (1S whoop AIO)

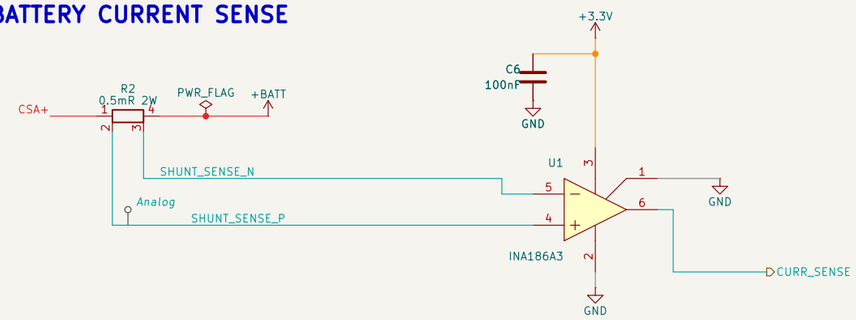
Size: A3	Date: 2026–10–09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 1/14

BATTERY INPUT



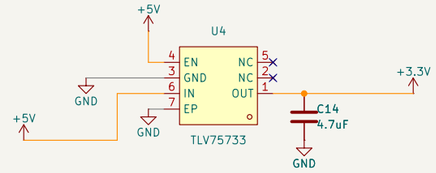
CSA+ = J2 (+) node before the shunt, J3 (−) = GND. 1S 3.0–4.35 V. PWR_FLAGs because the pads are passive. No reverse FET, no TVS.
Place the shared battery bulk close to the input pads.

BATTERY CURRENT SENSE



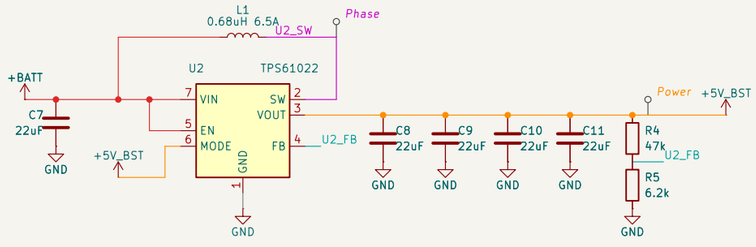
R2 0.5 mOhm Kelvin land: route SHUNT_SENSE_P/N as a pair on L1 or L6 (Analog class); every load taps +BATT after R2.
INA186A3 gain 100 = 50 mV/A: place U1 next to R2.

3.3 V LDO FOR FC + RX



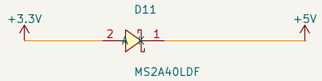
TLV75733 U4: EN on +5V; GND pin 3 and EP pin 7 grounded; NC pins 2/5. C13 10 uF input, C14 4.7 uF local output.
U4 permits 1–200 uF COUT; D11 bypasses reverse current during input collapse. Keep the bypass loop short.

5 V BOOST TPS61022



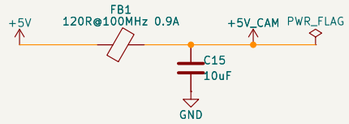
TPS61022: VOUT = 0.6 V x (1 + 47k / 6.2k) = 5.15 V, EN on +BATT, forced PWM, 4x 22 uF (C8–C11), 1.5 A continuous.
C8–C11 at pins 3/1, C7 at 7/1, minimal SW copper, R4/R5 at FB away from SW.

LDO REVERSE–CURRENT BYPASS



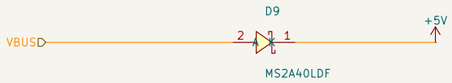
D11 anode on +3.3V / U4 OUT, cathode on +5V / U4 IN. TI TLV757P Fig 7–3 reverse–current bypass.
Place beside U4 with a short, wide loop. D9/D10 isolate the upstream USB and boost sources.
This bypass protects U4 when +5V falls before the +3.3V output capacitors discharge.

CAMERA 5 V FILTER

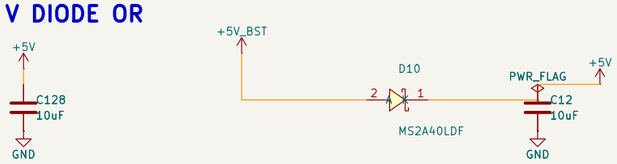


FB1 120 Ohm 0.9 A bead, C15 10 uF after it. +5V_CAM feeds J24 and J32. PWR_FLAG because FB1 is passive.

USB VBUS SCHOTTKY



5 V DIODE OR



5 V diode OR: D9 (VBUS) and D10 (+5V_BST) Schottky to +5V, the higher source wins. C12 10 uF at +5V.

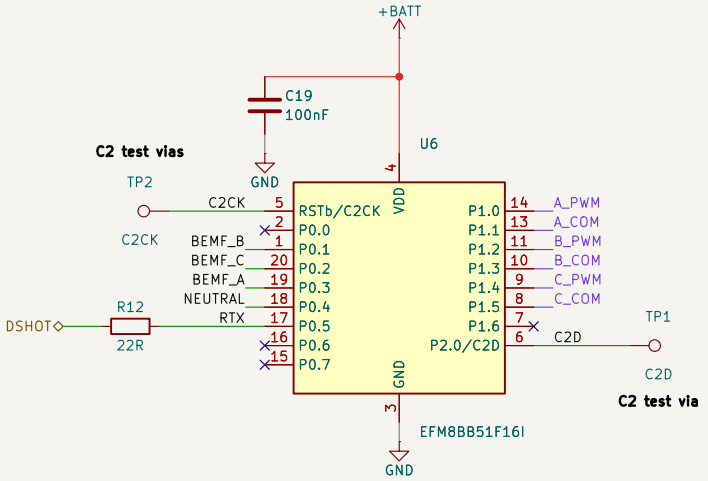
SHEET NOTES

Boost ground: U2 pin 1, C7 and C8–C11 GND form one island with no via inside the loop, joined to the planes by >= 4 vias at C7; keep the L2 plane solid below.
Keep it away from shunt, gyro, crystals, video, antennas.

OpenDrone

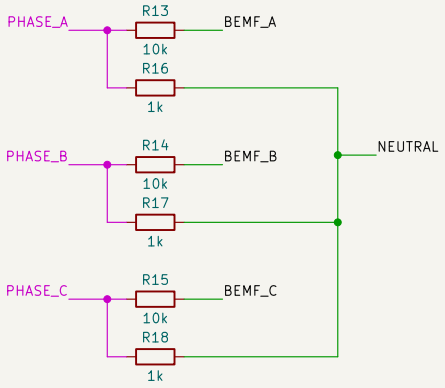
Sheet: /POWER/		
File: power.kicad_sch		
Title: Power: input, current sense, 5 V, 3.3 V, 04		
Size: A3	Date: 2026–10–09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 2/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



EFM8 pins per Bluejay BB51 layout A. VDD on +BATT; local 100nF directly at pin 4.
22R series damps DShot edges. U24 on the FC sheet translates and isolates domains. C2D/C2CK on test vias.

BEMF network: 10k series + 1k neutral star

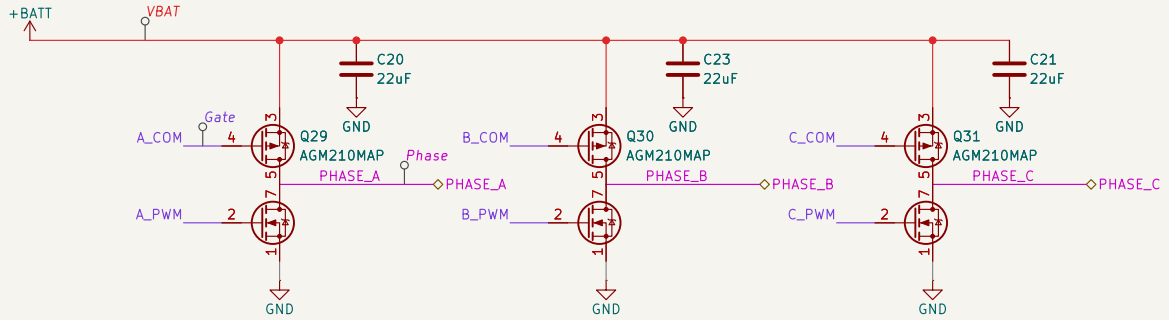


BEMF: 10k phase to comparator, 1k phase to neutral (P0.4). No GND leg at 1S.

ESC channel placed 4x (refs shown are ESC1):
ESC1 U6 M1 rear right, ESC2 U7 M2 front right, ESC3 U8 M3 rear left, ESC4 U9 M4 front left.

U24 translates bidirectional DShot between FC 3.3 V and the battery-powered EFM8.
Either supply absent: U24 isolates its ports; USB-only power does not drive the ESC supply.

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP P+N per phase: S1 GND, S2 +BATT, drains = PHASE_x (keep the island whole).
Hot loop: 22 uF 0603 per FET from S2 to S1 at the pins.

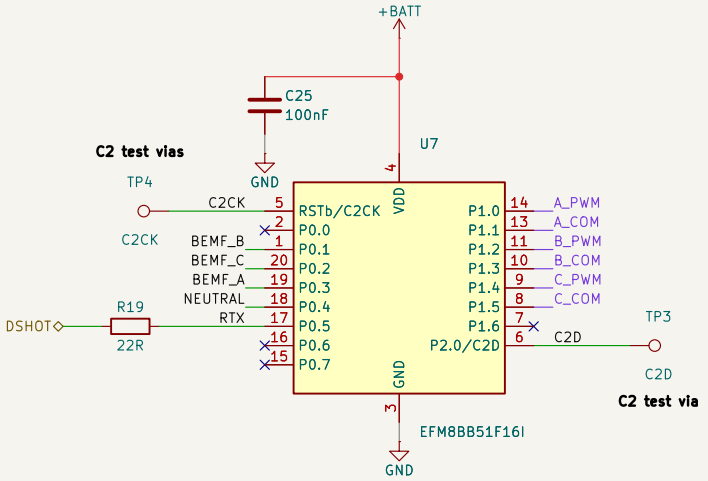
OpenDrone

Sheet: /ESC1/
File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

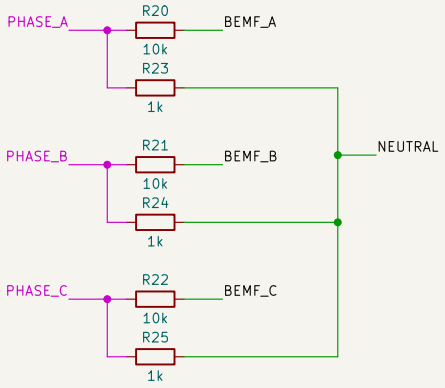
Size: A4	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 3/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



EFM8 pins per Bluejay BB51 layout A. VDD on +BATT; local 100nF directly at pin 4.
22R series damps DShot edges. U24 on the FC sheet translates and isolates domains. C2D/C2CK on test vias.

BEMF network: 10k series + 1k neutral star

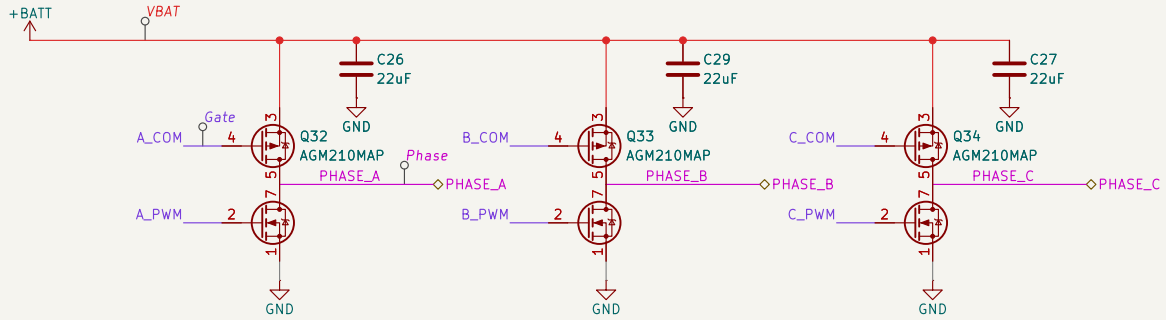


BEMF: 10k phase to comparator, 1k phase to neutral (P0.4). No GND leg at 1S.

ESC channel placed 4x (refs shown are ESC1):
ESC1 U6 M1 rear right, ESC2 U7 M2 front right, ESC3 U8 M3 rear left, ESC4 U9 M4 front left.

U24 translates bidirectional DShot between FC 3.3 V and the battery-powered EFM8.
Either supply absent: U24 isolates its ports; USB-only power does not drive the ESC supply.

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP P+N per phase: S1 GND, S2 +BATT, drains = PHASE_x (keep the island whole).
Hot loop: 22 uF 0603 per FET from S2 to S1 at the pins.

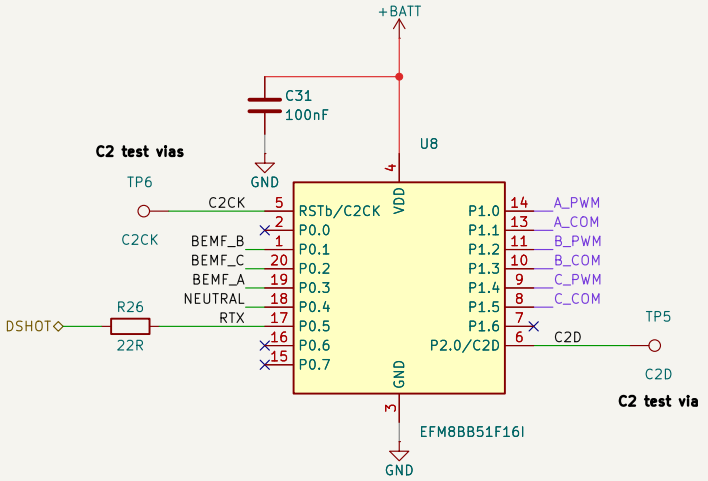
OpenDrone

Sheet: /ESC2/
File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

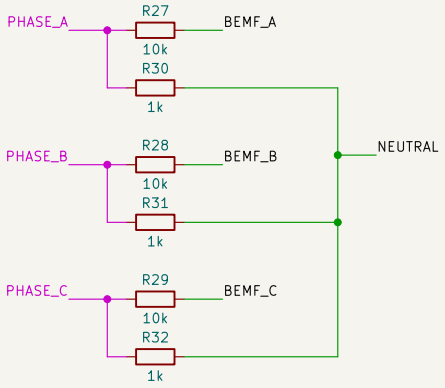
Size: A4	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 4/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



EFM8 pins per Bluejay BB51 layout A. VDD on +BATT; local 100nF directly at pin 4.
22R series damps DShot edges. U24 on the FC sheet translates and isolates domains. C2D/C2CK on test vias.

BEMF network: 10k series + 1k neutral star

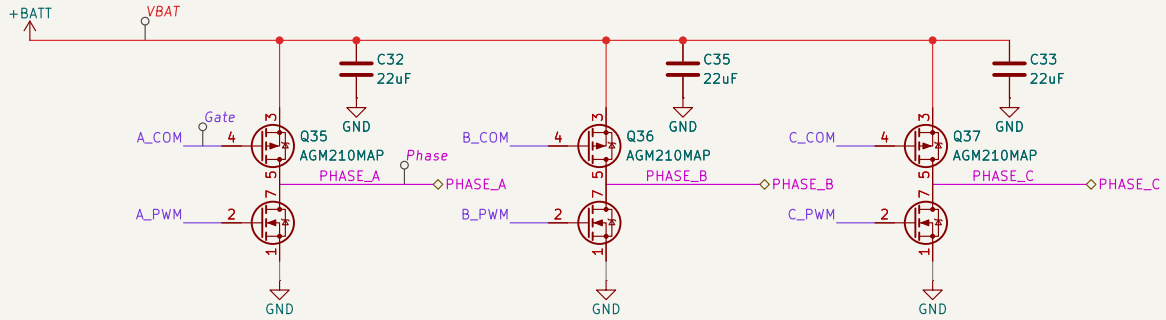


BEMF: 10k phase to comparator, 1k phase to neutral (P0.4). No GND leg at 1S.

ESC channel placed 4x (refs shown are ESC1):
ESC1 U6 M1 rear right, ESC2 U7 M2 front right, ESC3 U8 M3 rear left, ESC4 U9 M4 front left.

U24 translates bidirectional DShot between FC 3.3 V and the battery-powered EFM8.
Either supply absent: U24 isolates its ports; USB-only power does not drive the ESC supply.

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP P+N per phase: S1 GND, S2 +BATT, drains = PHASE_x (keep the island whole).
Hot loop: 22 uF 0603 per FET from S2 to S1 at the pins.

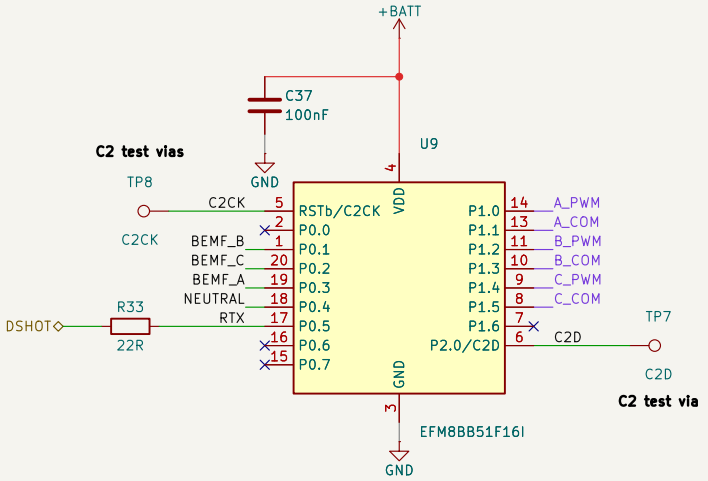
OpenDrone

Sheet: /ESC3/
File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

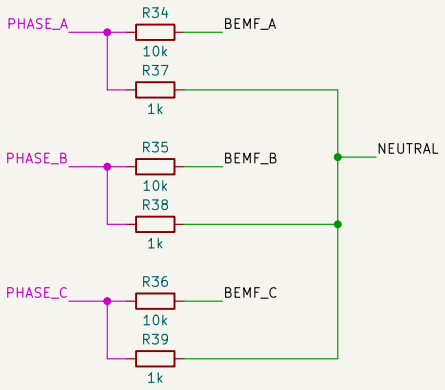
Size: A4	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 5/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



EFM8 pins per Bluejay BB51 layout A. VDD on +BATT; local 100nF directly at pin 4.
22R series damps DShot edges. U24 on the FC sheet translates and isolates domains. C2D/C2CK on test vias.

BEMF network: 10k series + 1k neutral star

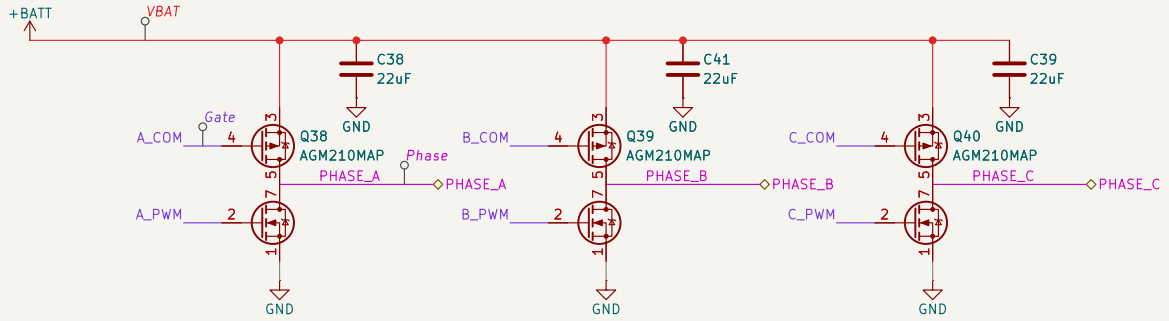


BEMF: 10k phase to comparator, 1k phase to neutral (P0.4). No GND leg at 1S.

ESC channel placed 4x (refs shown are ESC1):
ESC1 U6 M1 rear right, ESC2 U7 M2 front right, ESC3 U8 M3 rear left, ESC4 U9 M4 front left.

U24 translates bidirectional DShot between FC 3.3 V and the battery-powered EFM8.
Either supply absent: U24 isolates its ports; USB-only power does not drive the ESC supply.

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP P+N per phase: S1 GND, S2 +BATT, drains = PHASE_x (keep the island whole).
Hot loop: 22 uF 0603 per FET from S2 to S1 at the pins.

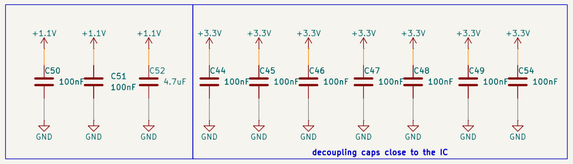
OpenDrone

Sheet: /ESC4/
File: esc_channel.kicad_sch

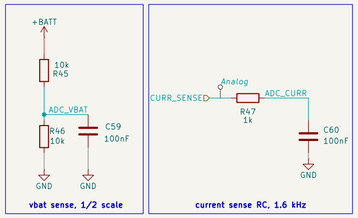
Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A4	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 6/14

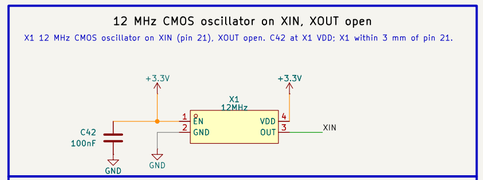
RP2354A: in-package flash

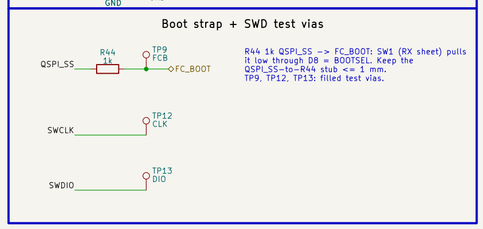


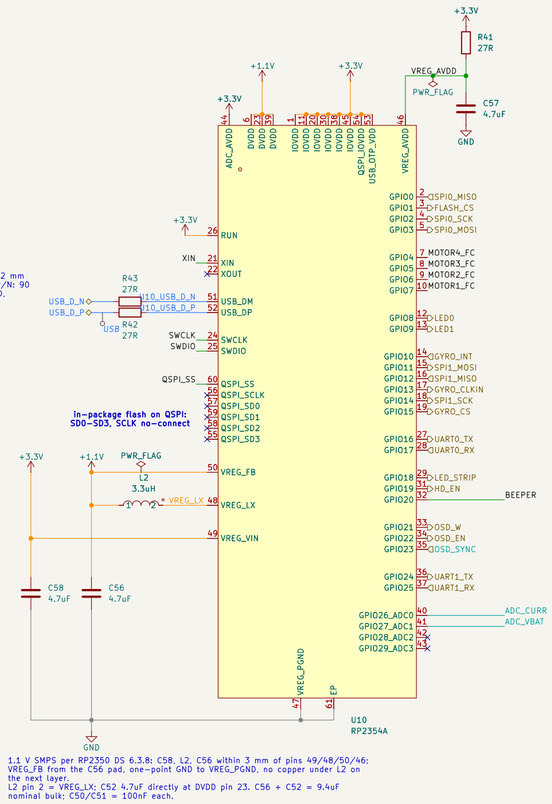
100 nF 0201 per supply pin: C49 at ADC_AVDD 44, C54 <= 1 mm from QSPI_IOVDD 54.



VBAT: R45/R46 10k/10k, 4.35 V -> 2.18 V on GPIO27. CURR: R47 1k + C60 100 nF on GPIO26.

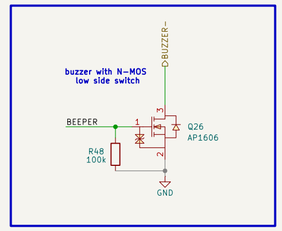




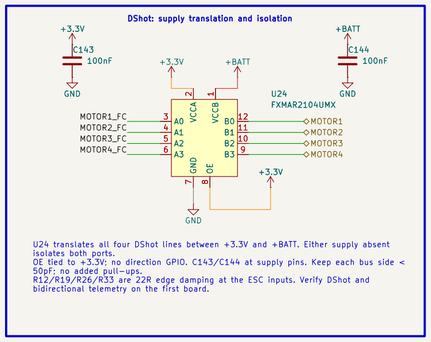


1.1 V SMPS per RP2350 DS 6.3.8: C58, L2, C56 within 3 mm of pins 49/48/50/46; VREG_FB from the C56 pad, one-point GND to VREG_PGND, no copper under L2 on the next layer.
L2 pin 2 = VREG_LX; C52 4.7uF directly at DVDD pin 23. C56 + C52 = 9.4uF nominal bulk; C50/C51 = 100nF each.

GPIO groups follow U10 at -90 deg: flash 0-3 east; motors 7/6/5/4 north to U24 A0-A3.
Gyro 10-15 west: INT10, MOSI11, MISO12, CLK13, SCK14, CS15.
Keep this order to match U11 pads.
UART0 16/17: strip18: HD_EN19: buzzer20. OSD21-23 and CRSF24/25 south. ADC26/27: spare28/29. LED±8/9.



Q26 AP1606 buzzer low-side switch: GPIO20 high = on. R48 100k holds it off through reset.



U24 translates all four DShot lines between +3.3V and +BATT. Either supply absent isolates both ports.
OE tied to +3.3V: no direction GPIO. C143/C144 at supply pins. Keep each bus side < 50pF; no added pull-ups.
R12/R19/R26/R33 are 22R edge damping at the ESC inputs. Verify DShot and bidirectional telemetry on the first board.

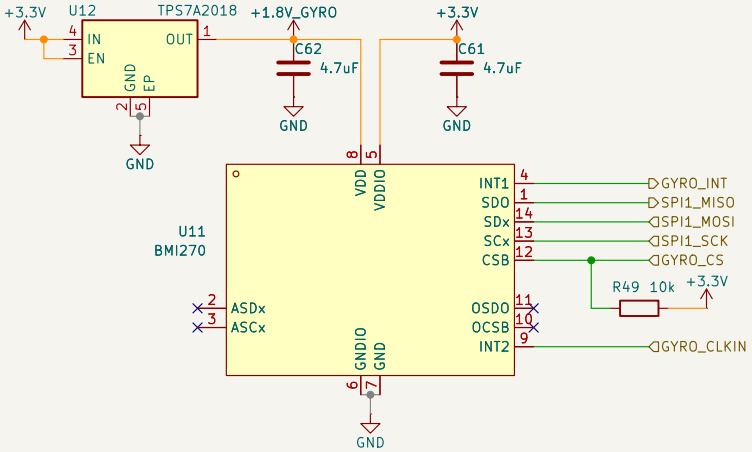
OpenDrone

Sheet: /RP2350A/
File: rp2350a.kicad_sch

Title: RP2354A flight controller MCU

Size: A2	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 7/14

IMU: BMI270 / ICM-42688-P LGA-14 land



BMI270 fitted: VDDIO +3.3V, VDD +1.8V_GYRO. ASDx/ASCx and OCSB/OSDO no-connect. R49 holds CSB high. ICM-42688-P uses the same land; requires its own decoupling population and firmware sensor/clock setup.



BMI270 local bypass: 100nF per supply pin. For ICM-42688-P: use 10nF on VDDIO and 100nF + 2.2uF on VDD; keep the LDO output capacitor.

U12: C61 at IN and C62 at OUT. C62 must retain >= 0.47uF effective capacitance.
C137 directly at U11 VDDIO pin 5; C138 directly at VDD pin 8. Short supply/ground loops to the adjacent plane.

OpenDrone

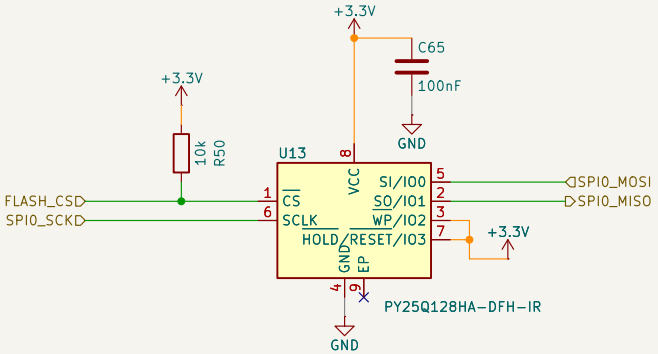
Sheet: /IMU/
File: imu.kicad_sch

Title: Gyro BMI270 + 1.8 V LDO

Size: A5	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 8/14

SPI NOR 16 MB BLACKBOX

WP# and HOLD# tied to +3.3V for single-SPI operation; pin 7 must not float.
R50 10k holds CS# high through reset. EP tabs no-connect.



Keep U13, SPI0 and FLASH_CS >= 3 mm from the RX antenna hole (SPI0 harmonics hit the ELRS band). C65 at pin 8.

OpenDrone

Sheet: /BLACKBOX/
File: blackbox.kicad_sch

Title: Blackbox 16 MB SPI NOR PY25Q128HA

Size: A5	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 9/14

VIDEO PATH

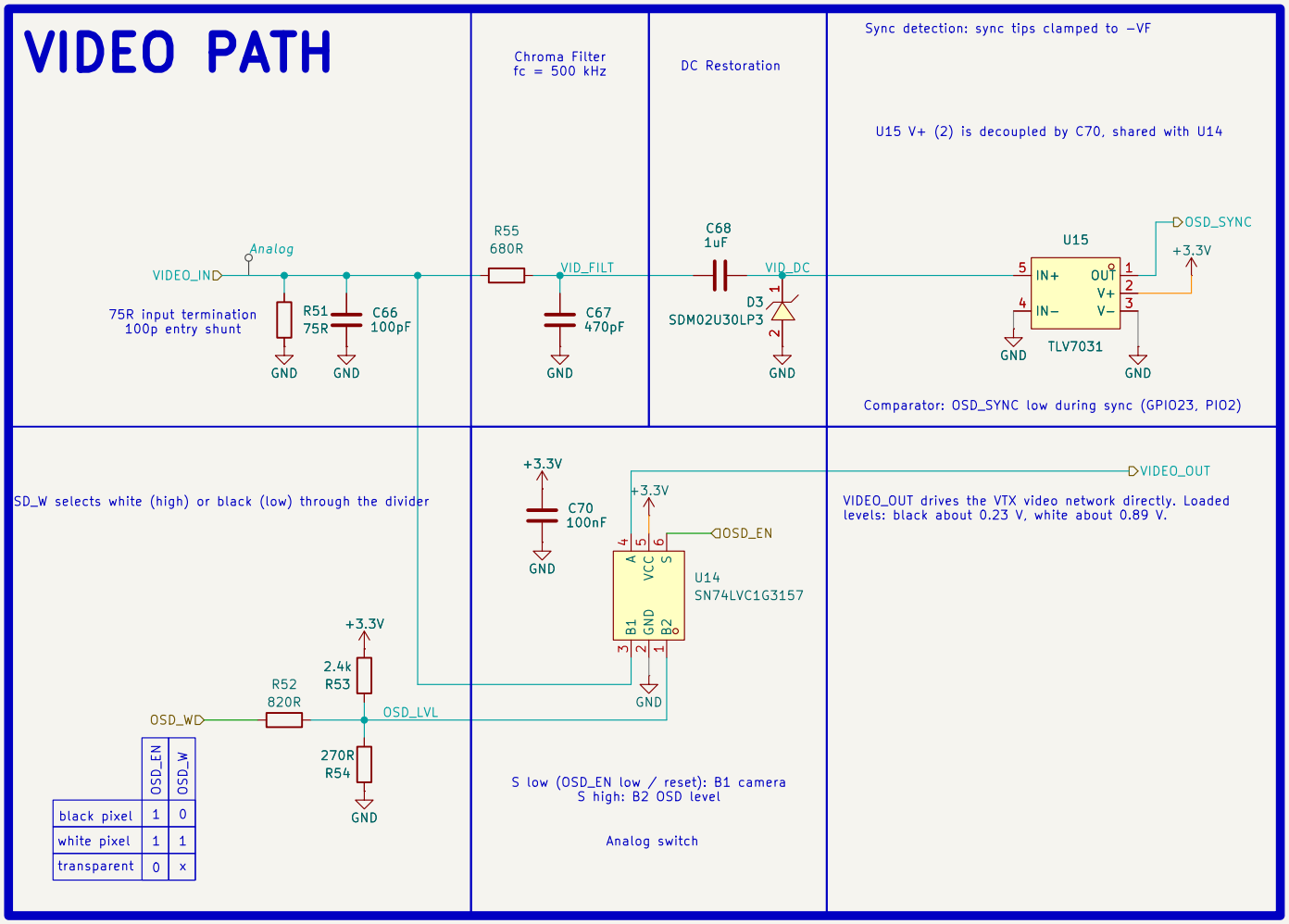
Chroma Filter
fc = 500 kHz

DC Restoration

Sync detection: sync tips clamped to −VF

U15 V+ (2) is decoupled by C70, shared with U14

OSD_EN low (reset): camera passes through. OSD_EN high: OSD_W high = white, low = black. No OSD without camera sync.
OSD_W/EN/SYNC = GPIO21/22/23 (PIO2).



	OSD_EN	OSD_W
black pixel	1	0
white pixel	1	1
transparent	0	x



C70 is the one supply cap of U14 and U15: place it between them. R51/C66 at the CAM pad line; video on L1/L6 over GND, away from the boost.

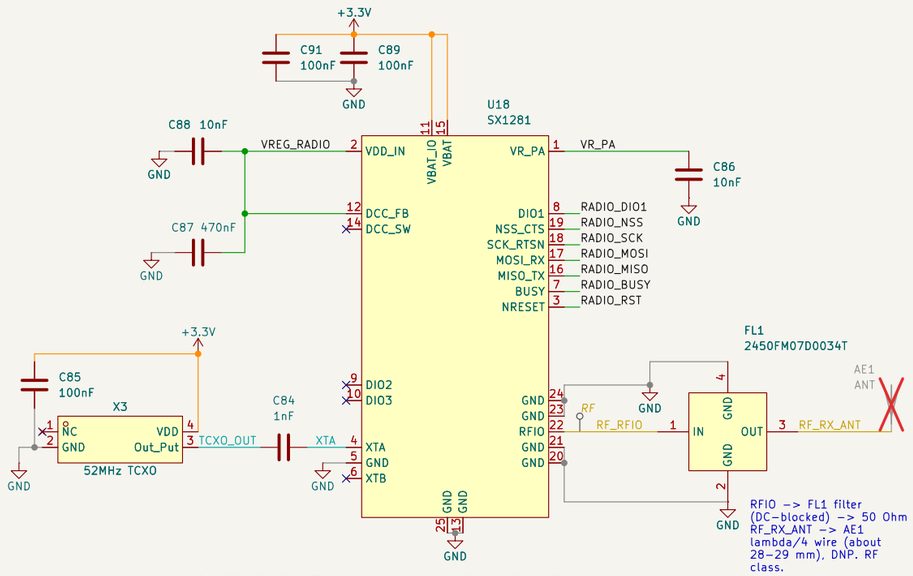
OpenDrone

Sheet: /OSD/
File: osd.kicad_sch

Title: PIO analog OSD front end

Size: A4	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 10/14

SX1281 RADIO + 2.4 GHz FILTER + WIRE ANTENNA

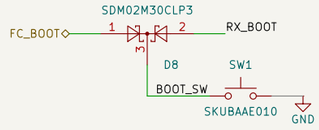


SX1281: LDO mode (DCC_SW open, no inductor), VREG_RADIO C87 + C88. Pin 5 is GND. 52 MHz TCXO X3 AC−coupled by C84 into XTA.

NOTES

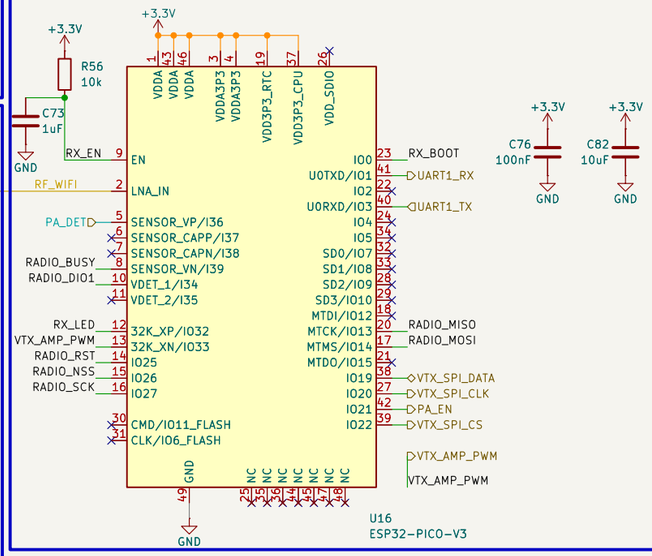
BOOT / BIND BUTTON

SW1 to GND, D8 dual Schottky to FC_BOOT and RX_BOOT; released, the diodes isolate the two straps. Held at power−up: RP2354A BOOTSEL + ESP32 download. Pressed while running: ELRS bind / Wi−Fi.

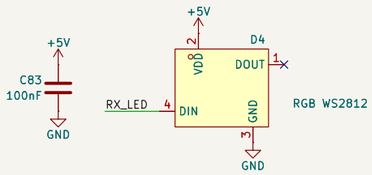


ESP32−PICO−V3 MCU (ELRS)

ESP32−PICO−V3: supplies on +3.3V, C82 + C76 at VDDA3P3; flash and crystal are in the SiP. EN = RX_EN with R56 10k + C73 1 uF.
GPIO0 = RX_BOOT, PA_EN GPIO21, PA_DET GPIO36. Wi−Fi TX peaks 370 mA.



ELRS RGB LED



D4 RGB LED (GPIO32): +5V supply, C83 100nF local bypass. Specified DIN VIH = 2.8V; direct 3.3V data is valid. Place C83 beside D4 VDD/GND.

WI−FI CHIP ANTENNA

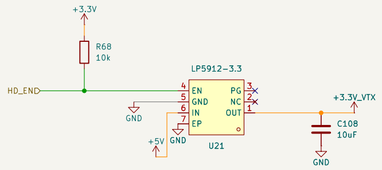


LNA_IN −> RF_WIFI 50 Ohm (<= 8 mm) −> pi match C133 (DNP) / R88 0R / C132 (DNP) −> AE2 chip antenna. Keep−out per DRU.

OpenDrone

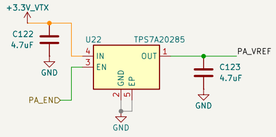
Sheet: /RX/		
File: rx_esp32_sx1281.kicad_sch		
Title: ELRS 2.4 GHz RX: ESP32−PICO−V3 + SX1281		
Size: A3	Date: 2026−10−09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 11/14

+3.3V_VTX LDO + HD GATE



U21 LP5912-3.3 on +5V, CIN = C13 on POWER if within 1 cm. EN = RP2354 GPIO19 (HD_EN).
High = analog enabled; low = HD / analog off. R68 10k holds analog on during reset. C108 at VCO supply.

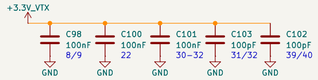
PA REFERENCE TPS7A20-2.85

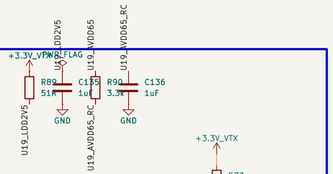


U22 TPS7A20285: 2.85 V +/-1.5% from +3.3V_VTX; EN = PA_EN (GPIO21), internal pull-down.
VIN >= 3.15 V; PA internal 2k load meets IOUT >= 1 mA. C122/C123 4.7 uF local bypass.

RTC6705 VTX

RTC6705 decoupling (pin numbers by each cap)

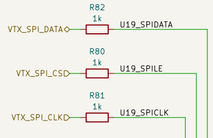




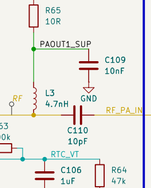
PNP PA DRIVE



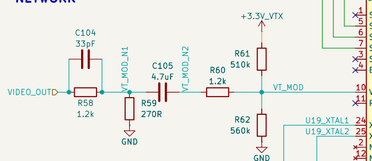
VTX SPI 1k SERIES

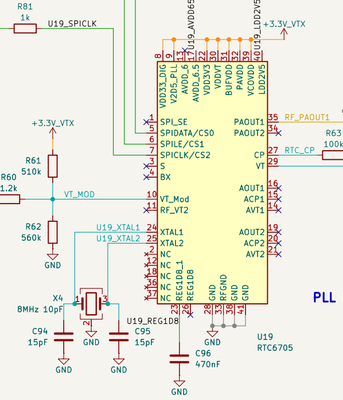


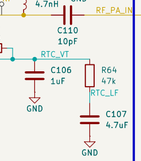
PAOUT1 FEED



VIDEO MODULATION NETWORK







PLL LOOP FILTER

RTC6705 pin 40 fed via R89 51R + C135 1u; pin 17 via R90 3.3k + C136 1u to GND. X4 8 MHz exactly, CL 10 pF.

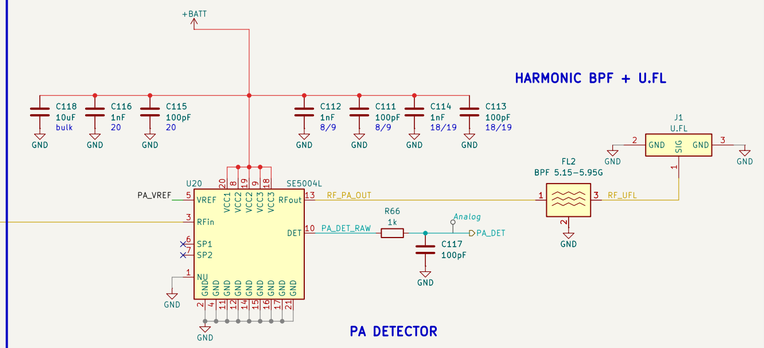
R80/R81/R82 1k in series on the SPI lines, placed at the ESP32 end: they limit back-feed into the unpowered RTC6705 in HD mode.

VIDEO_OUT -> R58 || C104, R59, C105, R60 -> VT_MOD with R61/R62 bias; loop filter CP -> R63 -> VT, C106, R64 + C107.
High-impedance (Analog class): short copper, >= 5 mm from boost and ESC phases.

ESP32 GPIO33 VTX_AMP_PWM -> 2 RC poles -> Q27 BC857 with emitter R73 -> PAOUT1 supply: sets the RTC6705 output-stage current.
R73 10R: at the top step R73/R65/Q27 run near derated ratings.

PAOUT1 feed: L3 4.7 nH choke from PAOUT1_SUP (C109). C110 10 pF DC block into the PA input. Only these between pin 35 and the PA: 50 Ohm CPWG < 5 mm, RF class.

SE5004L PA + SUPPLY, DETECTOR, BPF + U.FL



SE5004L (EK1 layout): RFout -> FL2 directly, no match parts. VCC on +BATT: decoupling at pins 8/9, 18/19, 20 + C118 10 uF. EP and GND pins to a via field.

PA_DET: DET -> R66 1k / C117 100p -> ESP32 GPIO36, RC at the ESP32 end. J1 U.FL with the L2 plane cut out under the signal pad, >= 4 GND vias.

SHEET NOTES

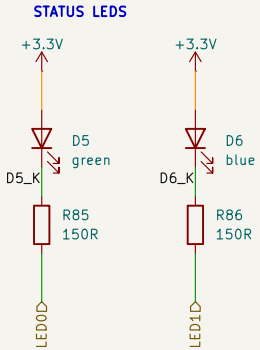
OpenDrone

Sheet: /VTX/
File: vtx.kicad_sch

Title: 5.8 GHz VTX: RTC6705 + SE5004L PA

Size: A2	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 12/14

+3.3V -> LED -> 150R -> GPIO, the pin sinks (low = on). LED0 GPIO8 green, LED1 GPIO9 blue, about 2–3 mA.
The LEDs may glow faintly during reset: cosmetic.



OpenDrone

Sheet: /LED/
File: led.kicad_sch

Title: Status LEDs

Size: A5	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 13/14

Battery pads (1S)

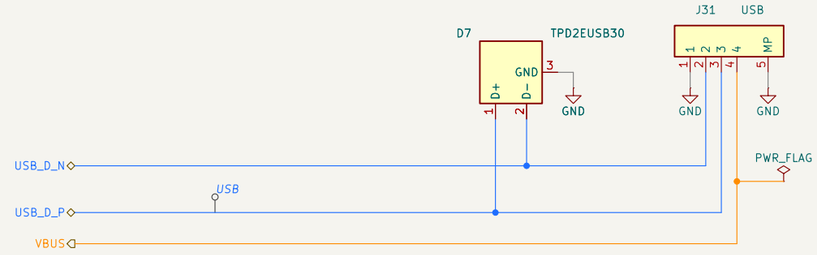
Battery pads J2 (+) / J3 (−) are drawn on POWER: 1.7 mm plated pads, >= 12 power vias each. 1S only, 3.0–4.35 V. Copper only (DNP).

MOTOR LANDS M1–M4



One PicoBlade land per motor, pads 1/2/3 = PHASE_C/B/A (order free, Bluejay sets direction). Top side, mouths facing out.
Solder the wires or fit the DNP header 53047-0310 (1.0 A contact, below motor current).

USB: JST SH 1.0 mm 4-pin + ESD



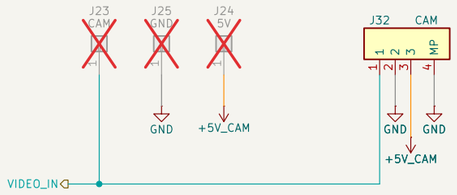
J31 JST SH: pin 1 GND, pin 2 D−, pin 3 D+, pin 4 VBUS; tabs to GND. Numbering follows JST drawing.
D7 ESD at J31. USB pair 90 Ohm, length-matched. Mating cable must match this pinout.

LED strip + buzzer row



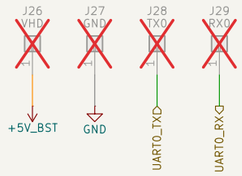
Row GND LED 5V BZ−. LED = LED_STRIP straight from GPIO18 (3.3 V logic). 5V feeds strip and buzzer; BZ− = drain of Q26. Copper only (DNP).

Camera pads + JST SUR 0.8 mm plug



J32 JST SUR: pin 1 VIDEO_IN, pin 2 GND, pin 3 +5V_CAM; tabs to GND. CAM / GND / 5V pads in parallel.
Numbering follows the JST drawing. Verify the mating camera cable follows this pinout.

HD pads (DJI O4 Lite)



VHD = +5V_BST, TX0/RX0 = UART0 GPIO16/17 for MSP DisplayPort. 3.3 V logic only.
Copper only (DNP).

Mounting holes



H1–H3: 25.5 mm M1.4 pattern cut in Edge.Cuts; no net, not in BOM or CPL.

PADS: every off-board connection. Solder pads are DNP and out of BOM / CPL.
Connector pin numbering verified against JST drawings; cable pinouts and motor-header heating need physical checks.

OpenDrone

Sheet: /PADS/		
File: pads.kicad_sch		
Title: Pads, motor lands, camera, HD, USB		
Size: A3	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 14/14